AI Engineer Roadmap

3 Months · 2 Hours/Day · 5 Days/Week · ~120 Hours Total

A structured path from software developer to production AI engineer by Md Rahmat Ullah (Head of AI Lab @ InfinitiBit GmbH)

About This Roadmap

This roadmap is designed for software developers who want to transition into AI engineering. It covers the full spectrum: mathematical foundations, classical machine learning, deep learning with PyTorch, large language models, RAG systems, AI agents, and production deployment — in exactly that order.

Each month contains 4 weeks of structured learning, with a milestone project at the end to reinforce and demonstrate your skills. By Month 3, you will be building and deploying production-grade AI services.

Daily	Days/Week	Monthly	Total
2 hours	**5 days**	**~40 hours**	**~120 hours**

MONTH 1 — FOUNDATIONS

Goal: Build the technical bedrock. Python fluency, maths intuition (linear algebra, calculus, statistics), and data wrangling — everything AI engineering sits on top of.

Week	Topics	Resources & Notes
Week 1 *Python for AI*	• Data types, loops, functions, OOP basics • List comprehensions, generators, decorators • File I/O, JSON, virtual environments • Type hints and basic debugging	• Udemy: Complete Python Bootcamp (Jose Portilla) • Python Official Documentation (docs.python.org) • Focus: write scripts daily, not just watch lectures
Week 2 *Linear Algebra*	• Vectors, matrices, dot products • Matrix multiplication (the backbone of neural nets) • Eigenvalues & PCA intuition • Implement all operations with NumPy	• FREE: 3Blue1Brown — Essence of Linear Algebra (YouTube) • FREE: NumPy Quickstart (numpy.org/learn) • Practice: implement matrix mul from scratch in Python
Week 3 *Calculus, Stats & Probability*	• Derivatives, chain rule, gradients • Probability distributions, Bayes' theorem	• FREE: 3Blue1Brown — Essence of Calculus (YouTube) • FREE: Khan Academy — Statistics & Probability

	• Mean, variance, standard deviation, covariance • Statistical intuition for ML evaluation	• Focus on intuition — you won't derive proofs on the job
Week 4 *Data Wrangling — Project*	• Pandas: DataFrames, groupby, merge, reshape • Data cleaning, handling missing values, outliers • Matplotlib & Seaborn visualisation • Exploratory Data Analysis (EDA) workflow	• Udemy: Data Analysis with Pandas & Python • Kaggle Datasets (kaggle.com) — use any tabular dataset • MILESTONE PROJECT: Full EDA report on a real dataset

Month 1 Milestone Project: Perform a full Exploratory Data Analysis (EDA) on a real-world Kaggle dataset. Clean the data, handle missing values, visualise distributions and correlations, and write a summary of findings. This is the data foundation for everything that follows.

MONTH 2 — CORE AI/ML

Goal: Understand and build ML and deep learning models. Classical ML with Scikit-learn, neural networks from scratch, then PyTorch — the industry standard for AI research and production.

Week	Topics	Resources & Notes
Week 5 *Classical ML*	• Linear & logistic regression, decision trees • Random forests, SVMs, K-Nearest Neighbours • Train/val/test splits, cross-validation, metrics • Scikit-learn pipelines and model persistence	• Udemy: Machine Learning A-Z (Eremenko & de Ponteves) • FREE: Scikit-learn User Guide (scikit-learn.org) • Practice: train 3 different models on same dataset, compare
Week 6 *Neural Networks from Scratch*	• Perceptron, activation functions (ReLU, Sigmoid, Tanh) • Forward & backward pass computed manually • Gradient descent variants: SGD, Adam, RMSProp • Loss functions, overfitting, dropout, L2 regularisation	• FREE: Andrej Karpathy — Zero to Hero series (YouTube) • FREE: 3Blue1Brown — Neural Networks playlist • Build a tiny MLP from scratch in NumPy — no shortcuts
Week 7 *PyTorch + CNNs*	• Tensors, autograd, Dataset & DataLoader • Building models with nn.Module • Convolutional layers for image classification	• Udemy: PyTorch for Deep Learning (Daniel Bourke) • FREE: PyTorch Official Tutorials (pytorch.org/tutorials) • Practice: CIFAR-10 classification with pretrained backbone

	• Transfer learning with ResNet / EfficientNet	
Week 8 *NLP + Transformers — Project*	• Tokenisation, word embeddings, attention mechanism • Transformer architecture (encoder / decoder) • Hugging Face: BERT, GPT-2, fine-tuning basics • Text classification, named entity recognition	• FREE: HuggingFace NLP Course (huggingface.co/learn) — excellent • FREE: DeepLearning.ai — Transformers short course • MILESTONE PROJECT: Fine-tune a sentiment classifier on a custom dataset

Month 2 Milestone Project: Fine-tune a BERT-based sentiment classifier using HuggingFace Transformers on a custom dataset (e.g., product reviews, social media posts). Achieve at least 85% validation accuracy. Publish the model card to HuggingFace Hub.

MONTH 3 — APPLIED AI ENGINEERING

Goal: Build production-grade AI systems. LLM APIs, RAG pipelines, AI agents, prompt engineering, MLOps, and deploying AI services — the real work of an AI engineer.

Week	Topics	Resources & Notes
Week 9 *LLMs + Prompt Engineering*	• OpenAI, Anthropic, and Gemini API integration • System prompts, few-shot, chain-of-thought prompting • Structured outputs: JSON mode, tool / function calling • Token limits, cost estimation, rate limit handling	• FREE: Anthropic Prompt Engineering Guide (docs.anthropic.com) • FREE: OpenAI Prompt Engineering Guide • FREE: DeepLearning.ai — ChatGPT Prompt Engineering for Devs
Week 10 *RAG + Vector Databases*	• Retrieval-Augmented Generation (RAG) architecture • Embeddings: OpenAI, Sentence Transformers • Vector DBs: ChromaDB, Qdrant, Pinecone • Chunking strategies, hybrid search, reranking	• Udemy: LangChain Bootcamp (Activeloop) • FREE: LangChain RAG Tutorial (python.langchain.com) • FREE: ChromaDB & Qdrant documentation
Week 11 *AI Agents + Orchestration*	• Tool calling / function calling patterns • LangGraph, AutoGen, and CrewAI fundamentals • Agent memory, planning, and reflection loops	• FREE: LangGraph Docs (langchain-ai.github.io/langgraph) • FREE: DeepLearning.ai — AI Agents in LangGraph • FREE: AutoGen Documentation (microsoft.github.io/autogen)

	• Multi-agent coordination and error recovery	
Week 12 *MLOps + Capstone*	• FastAPI: wrap your model as a REST service • Docker: containerise the full AI app • Deploy on AWS (EC2/Lambda) or Render • Logging, monitoring, prompt versioning	• Udemy: MLOps Bootcamp • FREE: FastAPI Documentation, Docker Getting Started • CAPSTONE PROJECT: Production RAG AI service — fully deployed

Capstone Project: Build and deploy a full RAG-based AI service. The system should accept documents as input, index them in a vector database, expose a chat API via FastAPI, containerise with Docker, and deploy to AWS or Render. This is your portfolio centrepiece.

TOP RECOMMENDED COURSES

Course	Provider / Instructor	Month	Cost
Complete Python Bootcamp	Jose Portilla (Udemy)	1	Udemy Business
Essence of Linear Algebra	3Blue1Brown (YouTube)	1	**FREE**
Essence of Calculus	3Blue1Brown (YouTube)	1	**FREE**
Machine Learning A-Z	Eremenko & de Ponteves (Udemy)	2	Udemy Business
Zero to Hero	Andrej Karpathy (YouTube)	2	**FREE**
PyTorch for Deep Learning	Daniel Bourke (Udemy)	2	Udemy Business
HuggingFace NLP Course	HuggingFace (huggingface.co/learn)	2-3	**FREE**
DeepLearning.ai Short Courses	Andrew Ng's team	2-3	**FREE**
LangChain & Vector DBs	Activeloop (Udemy)	3	Udemy Business
AI Agents in LangGraph	DeepLearning.ai	3	**FREE**
MLOps Bootcamp	Various (Udemy)	3	Udemy Business

TIPS FOR SUCCESS

- Don't just watch — code along every session. Passive video watching is the most common failure mode.
- Commit your code to GitHub from Day 1. An active GitHub profile is worth more than any certificate to hiring managers.
- The 3 milestone projects are non-negotiable. Real projects beat certifications every time.

- Use Udemy Business for paid courses — it covers all the Udemy recommendations in this roadmap.
- Andrej Karpathy's Zero to Hero and the HuggingFace course are arguably the best free resources available — prioritise them.
- Your existing 6+ years of software development experience is a massive advantage in Month 3 (APIs, Docker, deployment). You'll move faster than most beginners.
- Month 3 content (RAG, agents, LangGraph, FastAPI) maps directly to production AI services you can build at InfinitiBit immediately.

After 3 months you will be able to:

- Design and deploy production RAG pipelines backed by vector databases
- Build and orchestrate multi-agent AI systems using LangGraph or AutoGen
- Fine-tune and serve open-source LLMs for domain-specific tasks
- Containerise and deploy AI microservices on AWS or similar cloud platforms
- Architect end-to-end AI products integrating LLM APIs into real applications